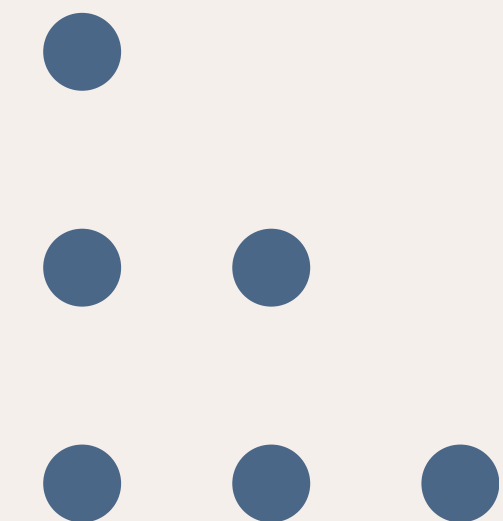




NOURISH SHARE

Real-Time Geospatial Matching Platform for
Urban Food Redistribution





Our Team

Mentor

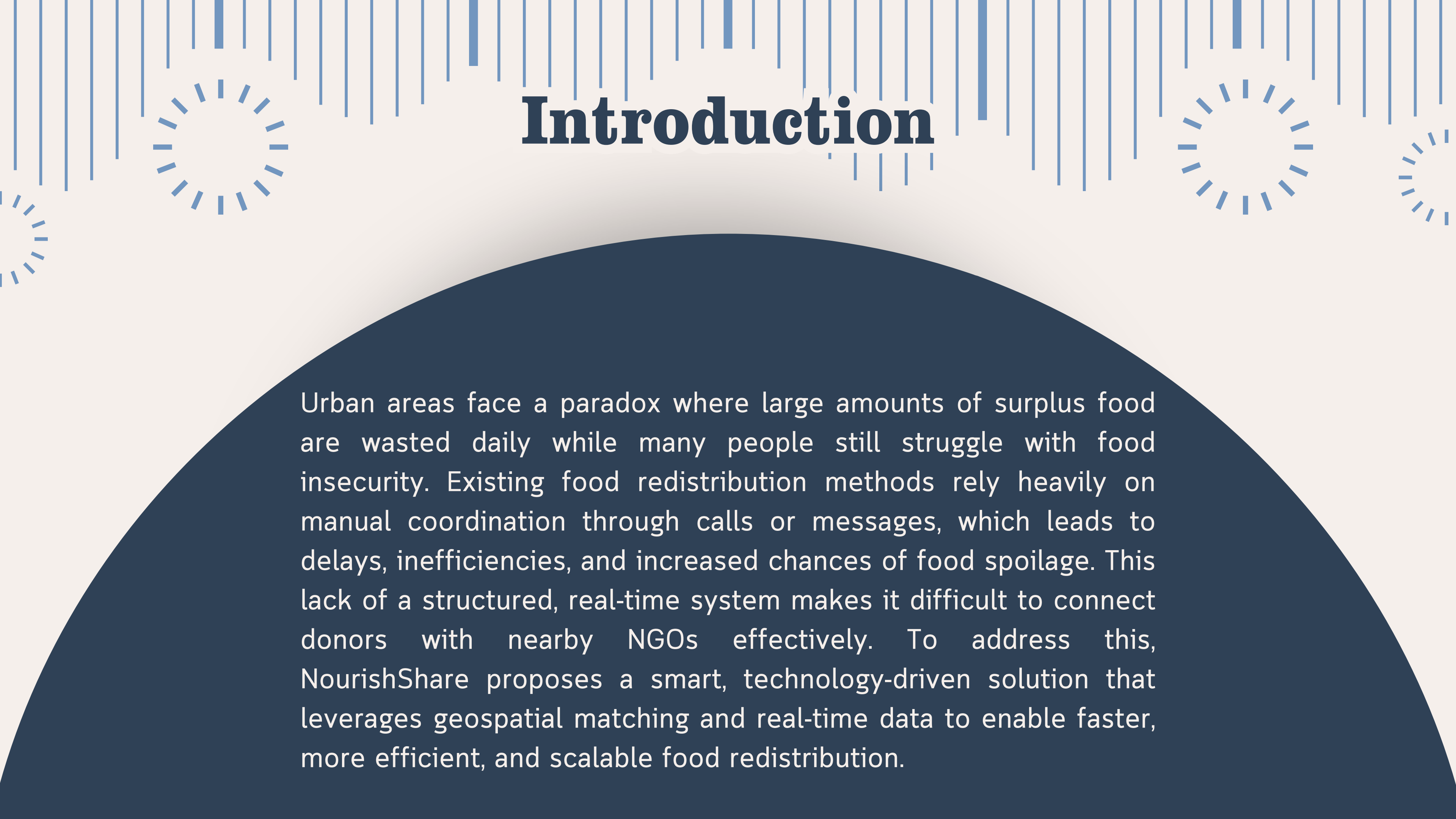
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Team Members



Introduction

Urban areas face a paradox where large amounts of surplus food are wasted daily while many people still struggle with food insecurity. Existing food redistribution methods rely heavily on manual coordination through calls or messages, which leads to delays, inefficiencies, and increased chances of food spoilage. This lack of a structured, real-time system makes it difficult to connect donors with nearby NGOs effectively. To address this, NourishShare proposes a smart, technology-driven solution that leverages geospatial matching and real-time data to enable faster, more efficient, and scalable food redistribution.

Objectives

- Develop a real-time geospatial matching system for food redistribution
- Reduce food wastage through efficient allocation
- Enable quick and automated coordination between donors and NGOs
- Implement distance and expiry-based prioritization for optimal matching
- Provide an interactive map interface for visualization and claiming
- Ensure secure user authentication and data handling
- Design a scalable system supporting multiple users simultaneously

Methodology

Data Input and Processing

Users register and log in, and donors submit food details like quantity, location, and expiry. The system converts addresses into coordinates using geocoding and stores the data for processing.

Geospatial Matching and Decision Logic

The system identifies nearby NGOs using location data and applies a scoring mechanism based on distance and expiry to select the best match.

Visualization and Execution

Matched results are shown on a map, allowing NGOs to view and claim donations. The system verifies claims and updates status in real time for smooth coordination.

Technology Stack

Core Development Stack (MERN)

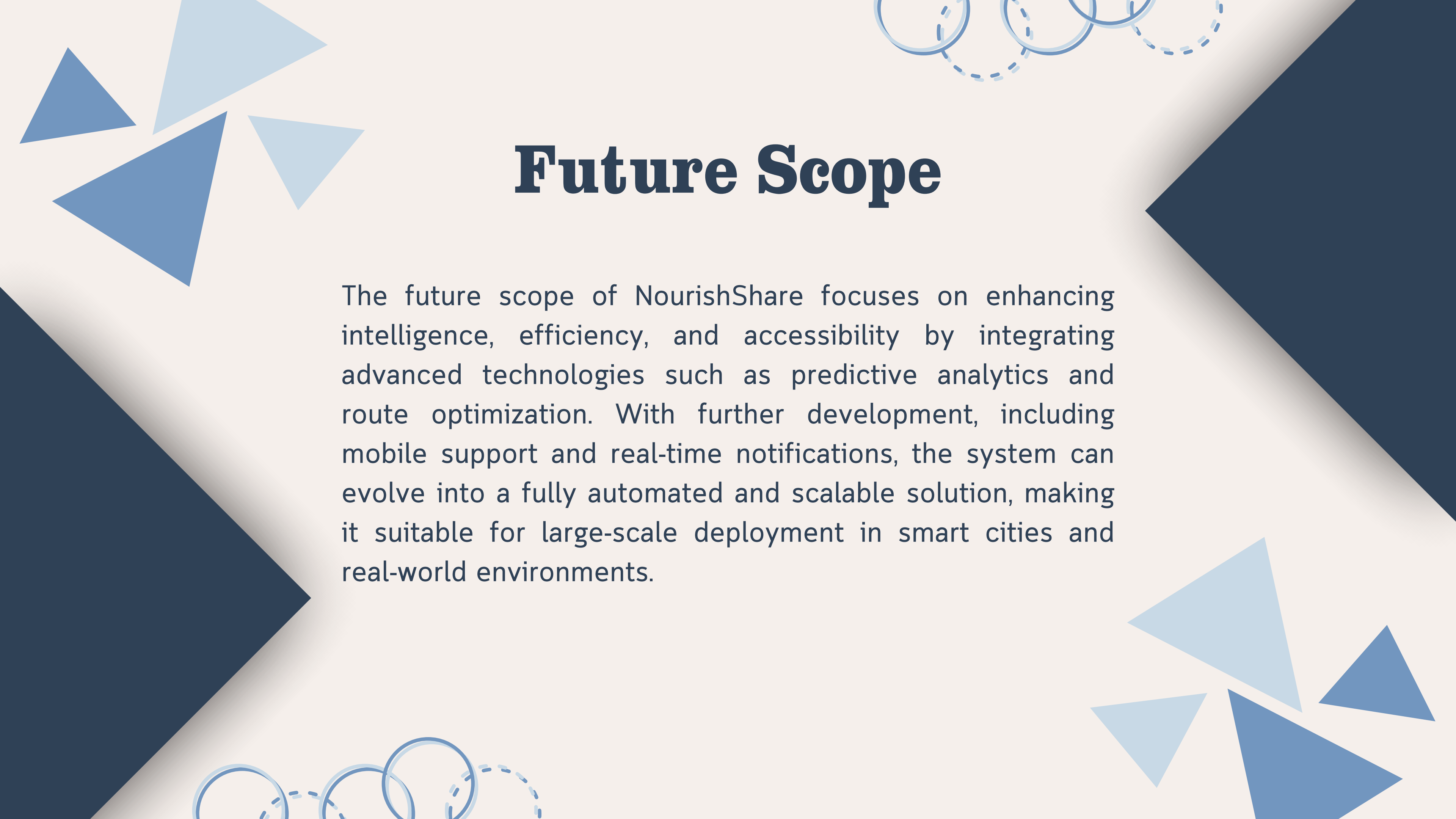
- MongoDB → NoSQL database
- Express.js → Backend framework
- React.js → Frontend
- Node.js → Server-side runtime for handling requests

Supporting Technologies & Services

- Google Maps API
- JWT (JSON Web Token)
- REST APIs
- Axios
- Tailwind CSS

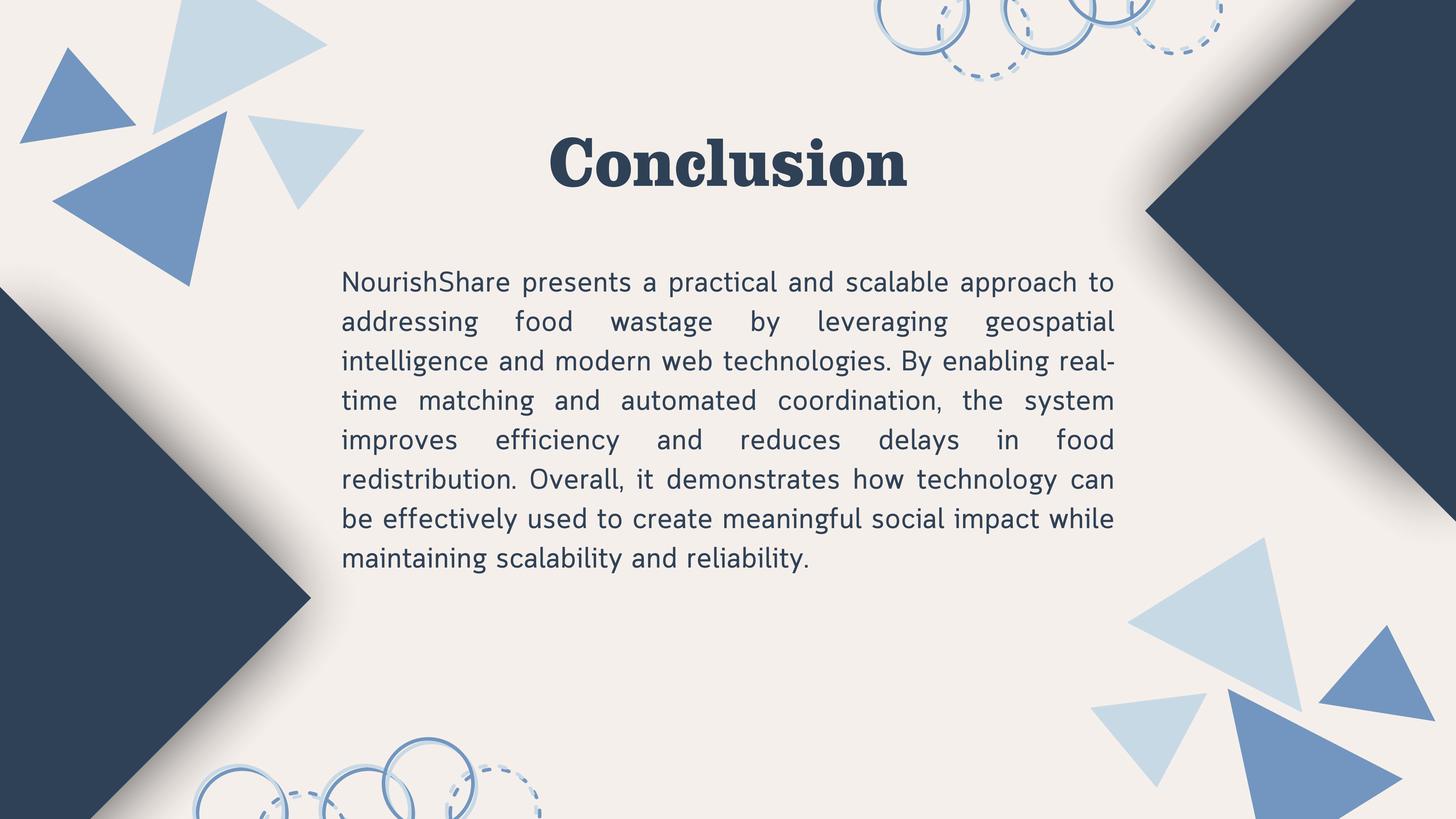
Results

- API Response Time: < 200 ms
- Efficient Matching: Reduced NGO search time via proximity filtering
- Stable Performance: Minimal delay under multi-user scenarios
- Accurate Allocation: Distance + expiry scoring improved match quality
- Reduced Food Wastage through timely redistribution
- Faster Coordination between donors and NGOs
- Lower Manual Effort via automated matching
- Scalable Solution for urban, high-demand environments
- Improved Transparency with real-time map visibility



Future Scope

The future scope of NourishShare focuses on enhancing intelligence, efficiency, and accessibility by integrating advanced technologies such as predictive analytics and route optimization. With further development, including mobile support and real-time notifications, the system can evolve into a fully automated and scalable solution, making it suitable for large-scale deployment in smart cities and real-world environments.



Conclusion

NourishShare presents a practical and scalable approach to addressing food wastage by leveraging geospatial intelligence and modern web technologies. By enabling real-time matching and automated coordination, the system improves efficiency and reduces delays in food redistribution. Overall, it demonstrates how technology can be effectively used to create meaningful social impact while maintaining scalability and reliability.



THANK
YOU!